

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Medium

Question

x	f(x)
0	5
1	$\frac{5}{2}$
2	$\frac{5}{4}$
3	$\frac{5}{8}$

The table above gives the values of the function f for some values of x. Which of the following equations could define f ?

Answer

- A. $f(x) = 5(2^x + 1)$
- B. $f(x) = 5(2^x)$
- C. $f(x) = 5(2^{-(x+1)})$
- D. $f(x) = 5(2^{-x})$

Correct Answer: D

Rationale

Choice D is correct. Each choice has a function with coefficient 5 and base 2, so the exponents must be analyzed. When the input value of x increases, the output value of f(x) decreases, so the exponent must be negative. An exponent of -x yields the values in the table: $5 = 5(2^{-0})$,

$\frac{5}{2} = 5(2^{-1})$, $\frac{5}{4} = 5(2^{-2})$, and $\frac{5}{8} = 5(2^{-3})$.

Choices A and B are incorrect and may result from choosing equations that yield an increasing, rather than decreasing, output value of f(x) when the input value of x increases. Choice C is incorrect and may result from choosing an equation that doesn't yield the values in the table.